EXPERIMENT - 68

1. Problem Statement

Write a recursive function to **convert a decimal number into its binary representation**.

2. Algorithm

Main Program

Step 1: START
Step 2: Read the decimal number n.
Step 3: Call decimal_to_binary(n).
Step 4: STOP

Function decimal_to_binary(n)

Step 1: If n / 2 is greater than 0, call decimal_to_binary(n / 2).
Step 2: Display n % 2.

3. Test Case Table

#	Decimal Number	Expected Output	Actual Output	Result
1	0	Binary: 0	Binary: 0	**PASS**
2	1	Binary: 1	Binary: 1	**PASS**
3	10	Binary: 1010	Binary: 1010	**PASS**
4	37	Binary: 100101	Binary: 100101	**PASS**
5	255	Binary: 11111111	Binary: 11111111	**PASS**
6	1024	Binary: 10000000000	Binary: 10000000000	**PASS**

4. Source Code

binary68.c

```
/* Experiment 68: decimal to binary using recursion. */
#include <stdio.h>

void decimal_to_binary(int n);

int main(void)
{
    int n;

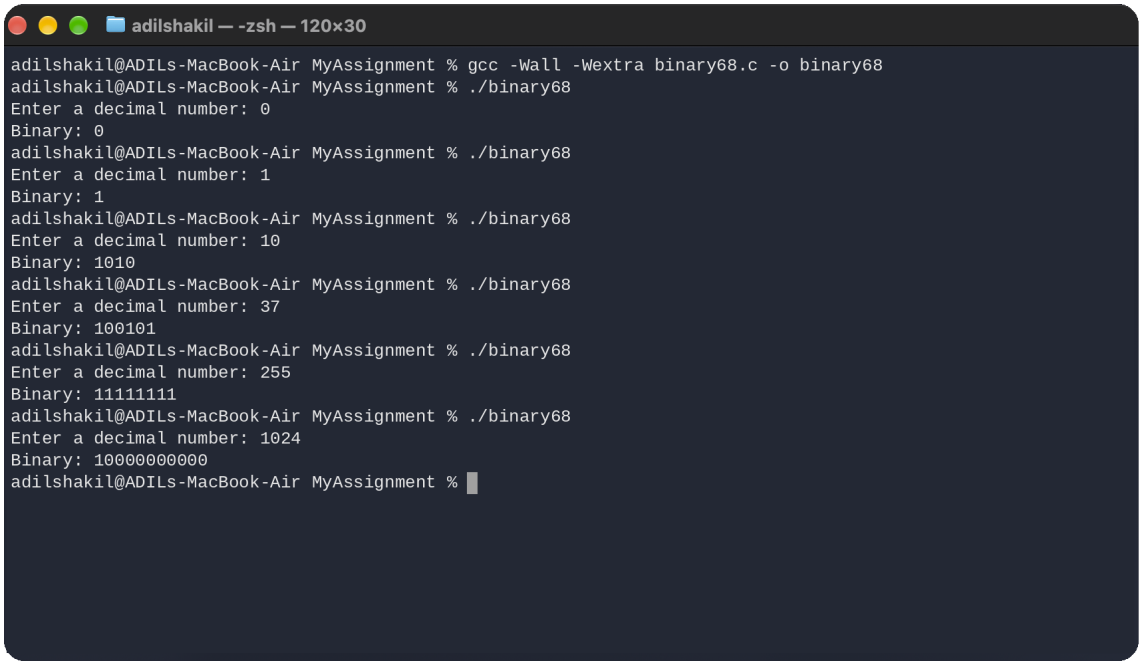
    printf("Enter a decimal number: ");
    scanf("%d", &n);

    printf("Binary: ");
    decimal_to_binary(n);
    printf("\n");
    return 0;
}

/* Convert n / 2 first, then print the remainder of n / 2.
   The remainders come out from the left bit to the right bit. */
void decimal_to_binary(int n)
{
    if (n / 2 > 0)
        decimal_to_binary(n / 2);
    printf("%d", n % 2);
}
```

5. Compilation and Output

Compiled with `gcc -Wall -Wextra binary68.c -o binary68` so the small warnings would show up too. It built clean.



All six passed.